Characterizing the academic OSPO in 2026

Overview

Here I am getting some background on the size and status of surveyed OSPOs.

My research questions:
- Where are university OSPOs located within their university?
- What are the OSPOs called? - How many of the organizations have an OSPO or dedicated OSPO staff? How many organizations just have staff doing OSPO-like work as part of another role?
- What fraction of OSPO contributors are volunteers?
- How old are the OSPOs?

Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(docstring)
```

```
Attaching package: 'docstring'

The following object is masked from 'package:utils':

    ?
```

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.4.3
```

```
source("utils.R")

# Read config.yml
cfg <- config::get()
```

Load data

```
ospo_loc <- read_tsv(file.path(cfg$data_path, "ospo_location.tsv"), show_col_types = FALSE, s
sata_other <- read_tsv(file.path(cfg$data_path, "sata_other.tsv"), show_col_types = FALSE, sh
qual <- read_tsv(file.path(cfg$data_path, "qual.tsv"), show_col_types = FALSE, skip_empty_rou
org_responses <- read_excel(file.path(cfg$curated_path, "orgs.xlsx"), sheet="responses")
mc_complex <- read_tsv(file.path(cfg$data_path, "mc_complex.tsv"), show_col_types = FALSE, sh
mc_other <- read_tsv(file.path(cfg$data_path, "mc_other.tsv"), show_col_types = FALSE, skip_c
mc_simple <- read_tsv(file.path(cfg$data_path, "mc_simple.tsv"), show_col_types = FALSE, skip
```

Where are university OSPOs located within their university?

```
# Need to do some data cleanup: filter so it's one response per institution,
# Also remove answers that are purely from a network, not a single campus

# To do this, I will keep only the FIRST occurrence of each unique value in 'organization'
```

```r
# Get a summary table of items and their first row index
tmp <- org_responses |>
  mutate(row_idx = row_number()) |>
  group_by(organization) |>
  summarise(first_occurrence_idx = min(row_idx), .groups = "drop")

# Remove the UC OSPO Network
tmp <- tmp[tmp$organization != "University of California OSPO Network", ]

rows_to_keep <- sort(tmp$first_occurrence_idx)

# Now we can look at results with one response per organization
ospo_loc_filtered <- ospo_loc[rows_to_keep,]
```

```r
# Extract the "Yes" count for each ospo location
# This was a SATA question so total != # of institutions
table_list <- lapply(ospo_loc_filtered, table)
yes_counts <- sapply(table_list, function(x) x["Yes"])
yes_counts <- sort(yes_counts, decreasing = TRUE)
print(yes_counts)
```

```
            tto.Yes         library.Yes           admin.Yes   academic_dept.Yes
                  8                   7                   5                   4
     rse_group.Yes research_center.Yes    ai_institute.Yes             hpc.Yes
                  3                   3                   3                   2
         joint.Yes
                  2
```

```r
sum(yes_counts)
```

```
[1] 37
```

The number of "yes" responses is substantially higher than the number of institutions (37 locations for 26 orgs). That's higher than I would expect from only 2 institutions where the OSPO is jointly hosted.

```r
num_yes_per_row <- rowSums(ospo_loc_filtered=="Yes")
sum(num_yes_per_row > 1)
```

```
[1] 5
```

While only 2 respondents selected "joint", 5 respondents selected more than one home for their OSPO. One selected library, tto, hpc, research center, AND joint!

How many OSPOs have neither a TTO nor library connection?

```
sum(ospo_loc_filtered$library == "No" & ospo_loc_filtered$tto == "No")
```

```
[1] 12
```

What are the OSPOs called?

Export to a new file for qualitative coding analysis. Again, let's filter out duplicates so we just have one response per institution.

```
qual_filt <- qual[rows_to_keep,]
write_df_to_file(qual_filt$ospo_function, "ospo_names.txt", path_key = "coding_path")
```

How many of the organizations have an OSPO or dedicated OSPO staff?

```
mc_comp_filtered <- mc_complex[rows_to_keep,]
mc_other_filtered <- mc_other[rows_to_keep,]
table(mc_comp_filtered$function_mc)
```

```
                A formally established OSPO
                                         13
An emerging OSPO (in a setup or pilot phase)
                                          5
OSPO-like activities without a formal office
                                          5
                                      Other
                                          3
```

```
# Keep commented for privacy!!
# mc_other_filtered$function_other[!is.na(mc_other_filtered$function_other)]
```

I see that one person who selected "other" wasn't sure if they should respond on behalf of their campus or their OSPO Network. I'm going to say they should answer on behalf of the campus, which adds +1 to the "OSPO-like activities without a formal office" category. The two other people who said "other" both wrote in "TTO", so I would argue that those should really be "OSPO-like activities without a formal office" as well.

Staff time

```
mc_simp_filtered <- mc_simple[rows_to_keep,]
time_counts <- table(mc_simp_filtered$staff_time)
desired_order <- c("Less than 0.5 FTE", "Between 0.5 and 1 FTE", "1 to 2 FTE", "2 to 5 FTE",
time_counts <- time_counts[desired_order]
time_counts
```

```
    Less than 0.5 FTE Between 0.5 and 1 FTE            1 to 2 FTE
                    7                     6                     7
          2 to 5 FTE          I don't know       More than 5 FTE
                    3                     2                     1
```

About half of OSPOs have less than 1 FTE. Most OSPOs (>80%) have 2 FTE or less.

Quick aside: Two people said "I don't know" for staff time commitment. Let's see if anyone else from their org gave a more informative answer.

```
counted_orgs <- org_responses[rows_to_keep,]
idk_rows <- which(mc_simp_filtered$staff_time=="I don't know")

# Keep commented for privacy!!
# counted_orgs[idk_rows,]
```

Our sample size is small enough that I can just take a look and see that both of these "I don't know" answers came from institutions that had only one response, so there are no second opinions here.

I suspect that the "OSPOs" with several FTE may actually be TTOs where everybody does OSPO stuff. Let's do an eyeball check to see if the data support this.

```r
combined <- cbind(ospo_loc_filtered, mc_comp_filtered$function_mc, mc_simp_filtered$staff_ti
names(combined)[ncol(combined) - 1] <- "function_mc"
names(combined)[ncol(combined)] <- "staff_time"

# Check that the data frame does not look weird
head(combined)
```

```
  library admin tto it hpc rse_group academic_dept research_center ai_institute
1     Yes    No Yes No Yes        No            No              Yes           No
2      No    No  No No  No        No            No              Yes           No
3      No   Yes  No No  No        No           Yes               No           No
4      No    No Yes No  No        No            No               No           No
5      No    No Yes No  No        No            No               No           No
6     Yes    No  No No  No        No            No               No           No
  joint                                    function_mc            staff_time
1   Yes                   A formally established OSPO           1 to 2 FTE
2    No An emerging OSPO (in a setup or pilot phase)     Less than 0.5 FTE
3    No                   A formally established OSPO           1 to 2 FTE
4    No OSPO-like activities without a formal office Between 0.5 and 1 FTE
5    No An emerging OSPO (in a setup or pilot phase)     Less than 0.5 FTE
6    No                   A formally established OSPO           2 to 5 FTE
```

```r
# FTE size vs level of establishment
table_establishment <- table(combined$staff_time, combined$function_mc)
print(table_establishment)
```

```
                      A formally established OSPO
  1 to 2 FTE                                    5
  2 to 5 FTE                                    3
  Between 0.5 and 1 FTE                         3
  I don't know                                  0
  Less than 0.5 FTE                             2
  More than 5 FTE                               0

                      An emerging OSPO (in a setup or pilot phase)
  1 to 2 FTE                                                     0
  2 to 5 FTE                                                     0
  Between 0.5 and 1 FTE                                          0
  I don't know                                                   1
  Less than 0.5 FTE                                              4
```

```
  More than 5 FTE                                                    0

                       OSPO-like activities without a formal office Other
  1 to 2 FTE                                                       1     1
  2 to 5 FTE                                                       0     0
  Between 0.5 and 1 FTE                                            2     1
  I don't know                                                     0     1
  Less than 0.5 FTE                                                1     0
  More than 5 FTE                                                  1     0
```

Eyeballing suggests that the established OSPOs have 0.5-5 FTE, while emerging OSPOs are most likely to have less than 0.5 FTE. OSPO-like activities without a formal office run the gamut from well-staffed to poorly staffed.

```
# FTE size vs. TTO status
table_tto <- table(combined$staff_time, combined$tto)
print(table_tto)
```

```
                       No Yes
  1 to 2 FTE            6   1
  2 to 5 FTE            3   0
  Between 0.5 and 1 FTE 2   4
  I don't know          1   1
  Less than 0.5 FTE     5   2
  More than 5 FTE       1   0
```

Meh, maybe the TTO hypothesis is not that interesting. It looks like the responses are pretty diverse, though Between 0.5 and 1 FTE is most common.

Basic statistics

```
# Fisher's exact test: are FTE size and establishment related?
fisher.test(table(combined$staff_time, combined$function_mc), simulate.p.value = TRUE)
```

```
    Fisher's Exact Test for Count Data with simulated p-value (based on
    2000 replicates)
```

```
data:  table(combined$staff_time, combined$function_mc)
p-value = 0.07246
alternative hypothesis: two.sided
```

Suggests the trend between staff time and OSPO establishment is not significant. Let's also run a Wilcoxon test, not to p-hack, but just to view the data from another perspective. Wilcoxon is a non-parametric test for a difference between medians of two groups.

```r
# Convert categories to their numerical midpoint
fte_numeric <- ifelse(combined$staff_time == "Less than 0.5 FTE", 0.25,
                  ifelse(combined$staff_time == "Between 0.5 and 1 FTE", 0.75,
                  ifelse(combined$staff_time == "1 to 2 FTE", 1.5,
                  ifelse(combined$staff_time == "2 to 5 FTE", 3.5, NA))))

comb_fte_numeric <- cbind(combined, fte_numeric)

# Test if established OSPOs have a different median FTE
# Create a simple Yes/No for formal establishment
comb_fte_numeric$is_formal <- ifelse(comb_fte_numeric$function_mc == "A formally established

fte_vs_establishment <- comb_fte_numeric[, c("fte_numeric", "is_formal")]
head(fte_vs_establishment)
```

```
  fte_numeric is_formal
1        1.50       Yes
2        0.25        No
3        1.50       Yes
4        0.75        No
5        0.25        No
6        3.50       Yes
```

```r
wilcox.test(fte_numeric ~ is_formal, data = fte_vs_establishment)
```

```
Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
compute exact p-value with ties


    Wilcoxon rank sum test with continuity correction

data:  fte_numeric by is_formal
W = 30.5, p-value = 0.02838
alternative hypothesis: true location shift is not equal to 0
```

Well, this test says the medians are significantly different. So all together, we have some weak evidence suggesting there might be a trend.

Volunteer time

```
table(mc_simp_filtered$volunteers)
```

```
           1 person       2 to 5 people         6-10 people        I don't know
                  4                  10                   4                   2
More than 10 people                None
                  1                   5
```

OSPO age

```
age_counts <- table(mc_simp_filtered$age)
desired_order <- c("Less than 1 year", "1 to 2 years", "3 to 5 years", "More than 5 years")
age_counts <- age_counts[desired_order]
age_counts
```

```
 Less than 1 year      1 to 2 years      3 to 5 years More than 5 years
                4                 9                 6                 5
```

Try to get a ballpark average age.

```
midpoint_lookup <- c(
  "Less than 1 year" = 0.5,
  "1 to 2 years"     = 1.5,
  "3 to 5 years"     = 4.0,
  "More than 5 years" = 5.5 # arbitrarily chosen
)
age_midpoints <- unname(midpoint_lookup[mc_simp_filtered$age])
mean(age_midpoints, na.rm = TRUE)
```

```
[1] 2.791667
```